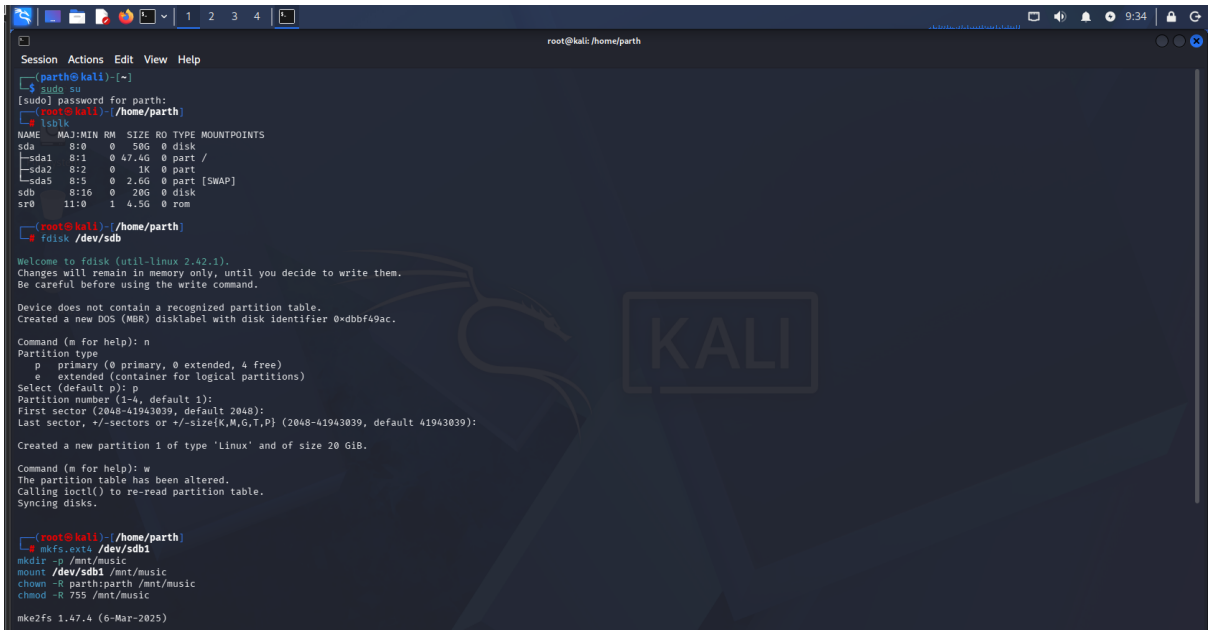
# ASSIGNMENT 3

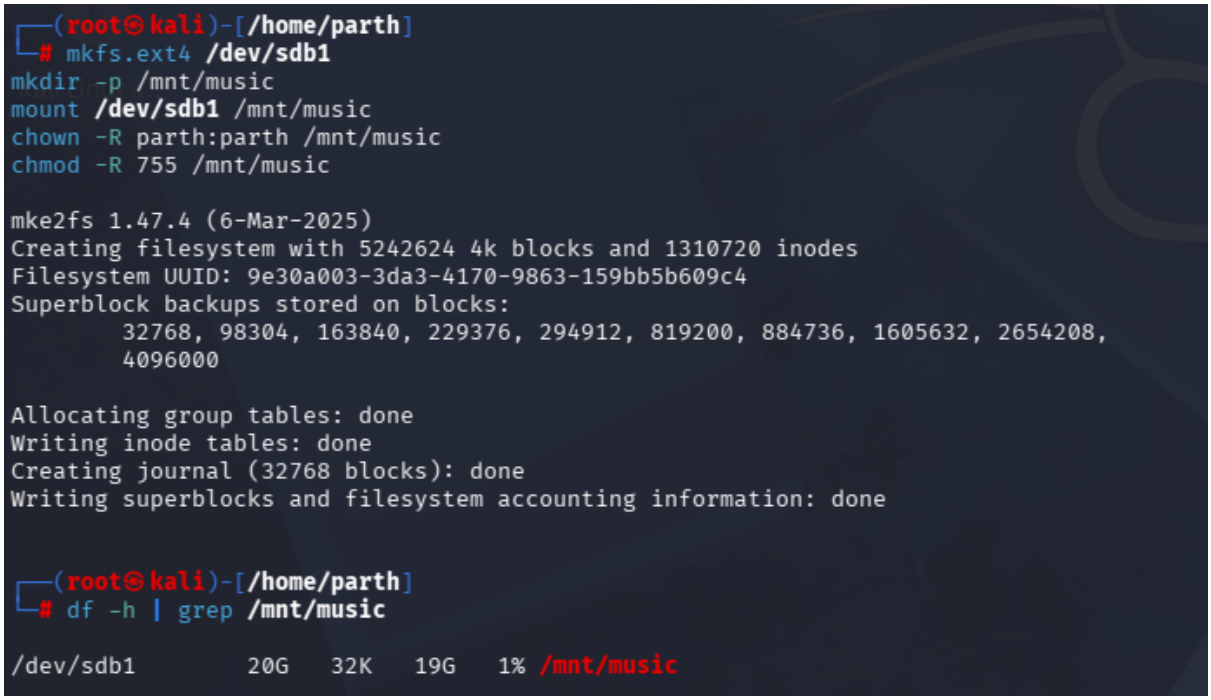
## Storage Management:

## Task 1.

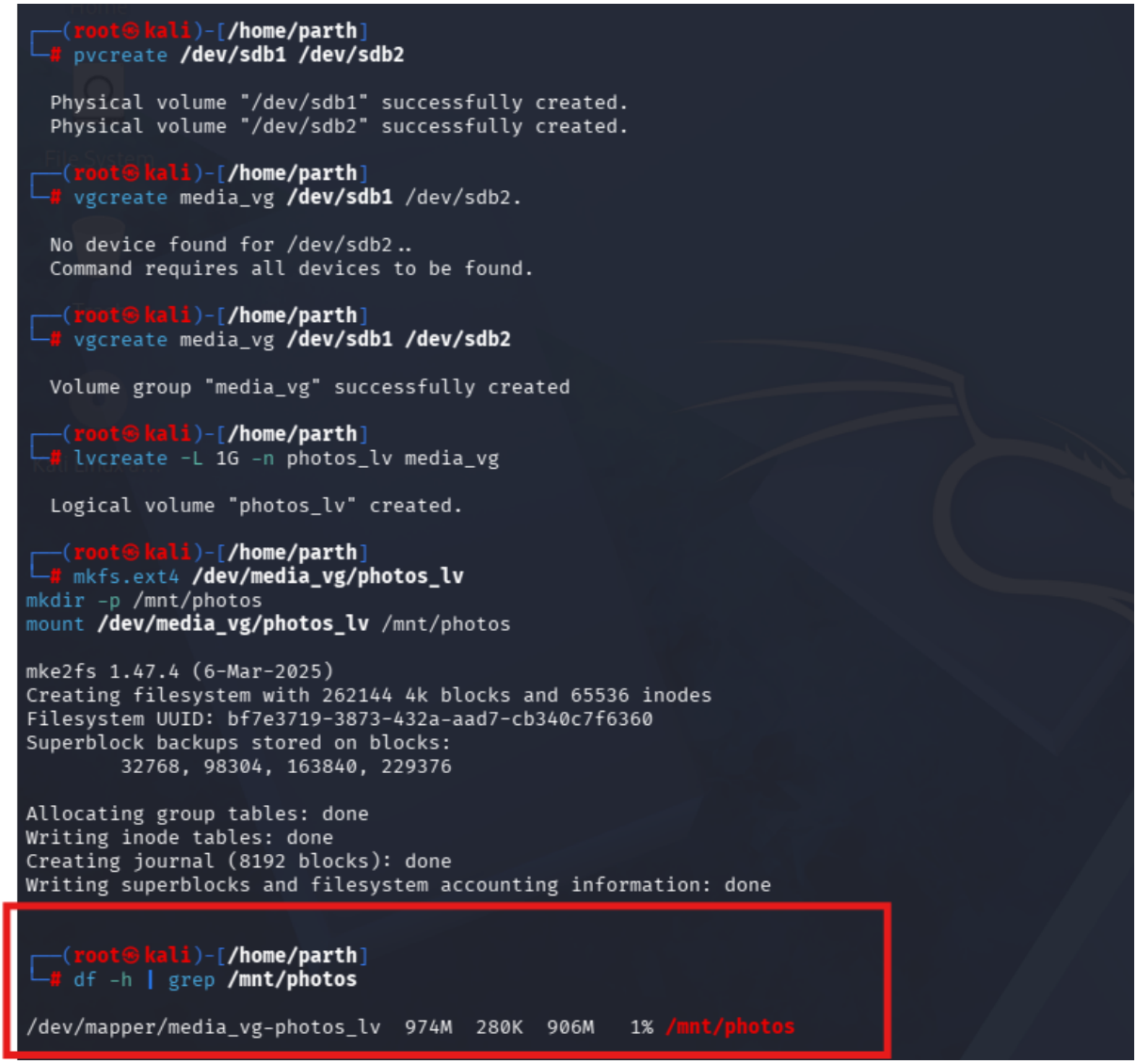
## Lsblk and fdisk formate process:



# After Moute Check:

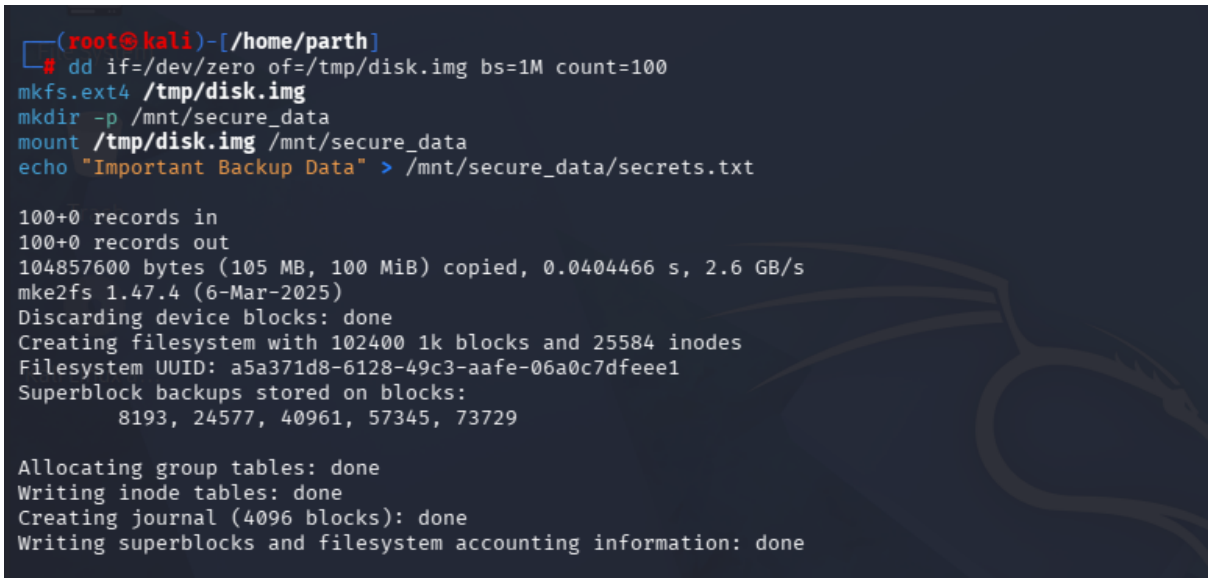


# Task 2.

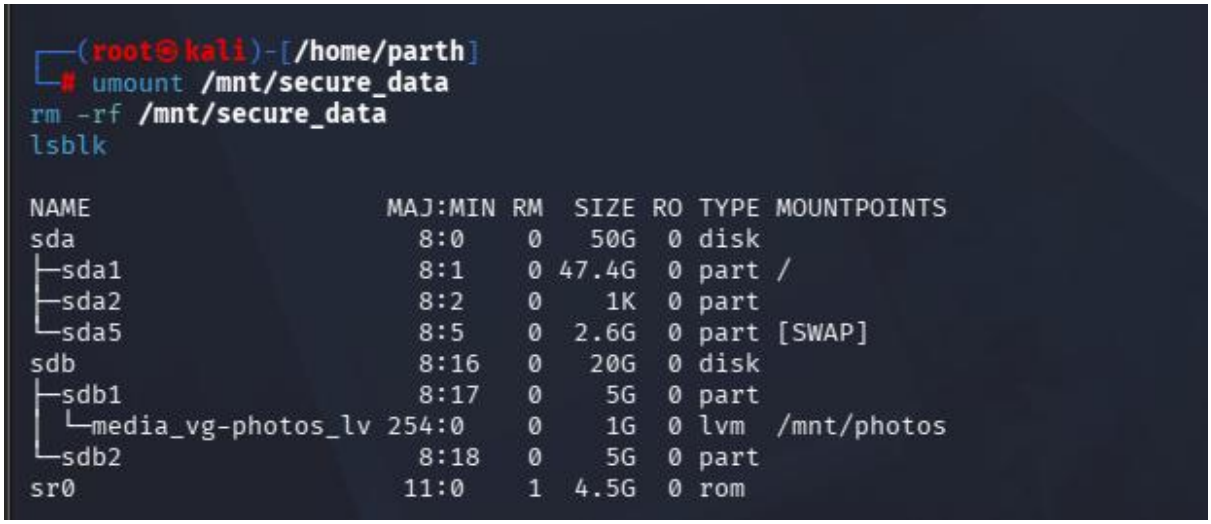

┌──(root㉿kali)-[/home/parth]
└─# pvcreate /dev/sdb1 /dev/sdb2
Physical volume "/dev/sdb1" successfully created.
Physical volume "/dev/sdb2" successfully created.
┌──(root㉿kali)-[/home/parth]
└─# vgcreate media_vg /dev/sdb1 /dev/sdb2.
No device found for /dev/sdb2..
Command requires all devices to be found.
┌──(root㉿kali)-[/home/parth]
└─# vgcreate media_vg /dev/sdb1 /dev/sdb2
Volume group "media_vg" successfully created
┌──(root㉿kali)-[/home/parth]
└─# lvcreate -L 1G -n photos_lv media_vg
Logical volume "photos_lv" created.
┌──(root㉿kali)-[/home/parth]
└─# mkfs.ext4 /dev/media_vg/photos_lv
mkdir -p /mnt/photos
mount /dev/media_vg/photos_lv /mnt/photos
mke2fs 1.47.4 (6-Mar-2025)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: bf7e3719-3873-432a-aad7-cb340c7f6360
Superblock backups stored on blocks:
32768, 98304, 163840, 229376
Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done
┌──(root㉿kali)-[/home/parth]
└─# df -h | grep /mnt/photos
/dev/mapper/media_vg-photos_lv 974M 280K 906M 1% /mnt/photos

# Task 3.

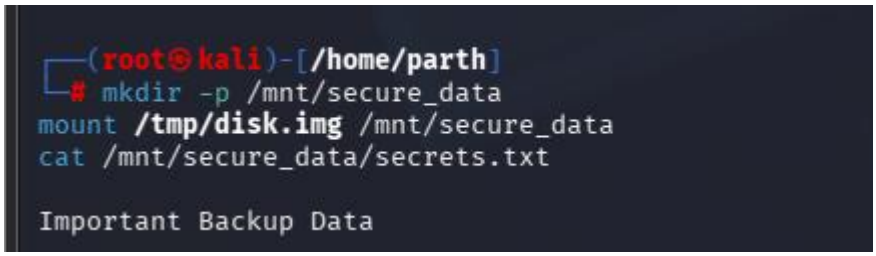
- **Step 1: Creating virtual disk and mounting the partition**




- **Step 2: Unmounting the partition and diagnosing with lsblk**



- **Step 3: Re-creating mount point and verifying recovered data**



# Task 4.

# Resize the file:



## Task 5.

### 1. dmesg | tail -n 20:

- **What it checks:** This command displays the latest messages from the kernel ring buffer. It tracks real-time hardware events registered by the operating system.
- **Troubleshooting purpose:** When a USB drive is physically inserted, the kernel immediately logs it. Running this command will show whether the system physically detected the connection. If successful, you will see lines containing text like New USB device found or usb-storage. If there is a hardware fault or bad port, it will log errors like I/O error or device not accepting address.

### 2. lsusb:

- **What it checks:** This command lists all USB controllers and physically connected USB devices discovered on the system's internal and external USB buses.
- **Troubleshooting purpose:** It helps isolate whether the issue is a detection problem or a mounting problem. If the vendor name of the USB drive (e.g., SanDisk, Kingston, Realtek) appears in this list, it means the operating system successfully detects the device chip at

the hardware subsystem level. If nothing appears, the issue points to a dead USB port, a faulty cable, or a broken USB drive.

## 3. lsblk:

- **What it checks:** This command lists details of all available block devices (such as hard drives, solid-state drives, and partition schemes) in a clear tree layout.
- **Troubleshooting purpose:** If lsusb shows that the hardware is recognized but it does not open on your desktop, lsblk will show you if the storage device layer actually mapped it (assigning names like /dev/sdb or /dev/sdc). By checking the **MOUNTPOINTS** column, you can immediately tell if the partition is recognized but simply failed to auto-mount, indicating a software configuration or filesystem issue rather than a hardware failure.